

An oxalato-bridged dimetallic heptanuclear $[K_5Fe_2^{III}]$ complex anion with guanidinium as counter cation: Synthesis, crystal structure, Hirshfeld surface analysis and magnetism

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ABSTRACT

An oxalato-bridged heterometallic heptanuclear $[K_5Fe_2^{III}]$ hybrid salt, $(CH_6N_3)[K_5(H_2O)_4Fe_2(C_2O_4)_6]$ (**1**) ($CH_6N_3^+$ = guanidinium cation) has been synthesized and characterized by elemental and thermal analyses, IR spectroscopy, single-crystal X-ray diffraction, EPR and SQUID magnetometry. The asymmetric unit in **1** is composed of a dimetallic heptanuclear $[K_5(H_2O)_4Fe_2(C_2O_4)_6]^-$ anionic complex and a guanidinium $CH_6N_3^+$ cation insuring the charge compensation. Each Fe^{3+} center in the anionic complex is six-coordinate in a distorted octahedron defined by three chelating oxalato(2-) ligands. In contrast, the coordination numbers for K^+ centers range from 7 to 9. The structural feature of focal interest in **1** is formation of an extended anionic network encapsulating guanidinium cations. This structure can be considered a sheet-MOF, where metal ions and oxalates (linkers) form infinite 2D units. The 3-D supramolecular architecture is stabilized by $O-H\cdots O$ and $N-H\cdots O$ hydrogen bonds linking polymeric oxalato-bridged anionic assemblies and organic cations. The thermal studies confirmed the anhydrous character of salt **1** and showed a three-step decomposition to yield a mixture of Fe_2O_3 and K_2O as final residue. EPR spectrum of **1** is in accordance with the oxidation state + 3 of the iron center in an octahedral environment. Temperature-dependence susceptibility measurements investigated in the temperature range 2–300 K revealed weak antiferromagnetic coupling at low temperatures in salt **1**. The distributions of different types of intermolecular interactions in the crystal structure were quantified by Hirshfeld surface analysis and their associated fingerprint plots.

1. Introduction

Organic-inorganic hybrid salts (OIHSs) represent a new class of crystalline materials formed by the self-assembly of both the flexible organic cations and inorganic counterparts. Currently, this class of materials combining multiple properties and functionalities has aroused great interest not only because of their versatile structural architectures [1,2] but also due to their wide applications in various fields including catalysis [3], molecular electronics and spintronics [4], metallic conduction [5], nonlinear optics [6,7] and magnetism [8–10]. Moreover, these materials are inclined to undergo phase transition under external temperature stimuli, making them potential candidates for synthesizing

dielectric switchable devices [11]. Among the OIHSs, oxalato-bridged complexes have been intensively studied over the last few decades. The sustained interest in these compounds could be ascribed to their rich structural diversity [12–14] and topology [15] along with special relevance in molecular magnetism [16–18]. Another aspect of paramount interest of oxalate compounds is CO_2 capture and storage, especially when porous materials are concerned [19].

As one of the simplest ligands, the oxalate dianion $C_2O_4^{2-}$ has attracted the attention of many research groups, owing to its ability to strongly bind metals in diverse connection modes such as monodentate or bidentate, forming various structures with zero-, one-, two-, or three-dimensional frameworks [20]. Furthermore, the remarkable ability of

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the oxalate ligand to mediate efficient magnetic coupling between adjacent magnetic centers has allowed the construction of many functional materials [21].

Compared to the very large number of homometallic oxalatometalate(III) hybrid salts [22–27], the heterometallic oxalato-bridged complexes have been much less explored [28,29]. Taking this into account, our research group has been involved in the development of heteropolynuclear anionic complexes involving pyridinium derivatives as counter cations [30], the ultimate goal being to gain some insight into the structural features and magnetic trends associated with this family of compounds. In continuation of this research work, we herein report the synthesis and characterization of a novel oxalato-bridged heterometallic heptanuclear $[K_5Fe_2^{III}]$ hybrid salt, $(CH_6N_3)[K_5(H_2O)_4Fe_2(C_2O_4)_6]$ (**1**) ($CH_6N_3^+ =$ guanidinium cation). The thermal behavior, EPR, magnetic properties and Hirshfeld surface analysis have also been investigated.

2. Experimental

2.1. Materials and physical measurements

All reagents for the synthesis were purchased from commercial sources and used as received. Elemental analysis for C, H and N was performed with a Perkin–Elmer 2400 analyzer. The infrared spectrum was recorded on a Perkin–Elmer 2000 spectrometer from KBr pellets in the range 4000–400 cm^{-1} . The powder diffraction pattern was obtained using a rotating anode SMARTLAB XE diffractometer from Rigaku company. The experiment was realized in reflection mode (Bragg–Brentano geometry), using $CuK\alpha$ radiation. The powder sample was obtained by hand single-crystals grinding and deposited in a low background Si sample holder. Thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC) were carried out on a LINSEIS STA PT-1000 thermoanalyzer instrument. The powdered sample (ca. 20 mg) was heated in air from 25 to 700 $^{\circ}C$ at a scan rate of 10 $^{\circ}C\ min^{-1}$. Electron paramagnetic resonance (EPR) measurements were performed with a Bruker ELEXYS E500 spectrometer operating at 9 GHz. The samples, each containing around 30 mg of loose powder, were placed into 4 mm diameter quartz tubes. The measurements were performed at room temperature with a microwave power of 0.3 mW and an amplitude modulation of 3G. Variable temperature magnetic susceptibilities were measured with a Quantum Design MPMS-5XL SQUID magnetometer. Measurements were done at 0.1 T down to 2 K on polycrystalline samples. Diamagnetic corrections were made with Pascal's constants for all the constituent atoms [31].

2.2. Synthesis of $(CH_6N_3)[K_5(H_2O)_4Fe_2(C_2O_4)_6]$ (**1**)

The precursor salt $K_3[Fe(C_2O_4)_3] \cdot 3H_2O$ was prepared by the direct combination of $FeCl_3$ and $K_2C_2O_4 \cdot H_2O$ solutions, as recently described in the literature [32]. A freshly prepared $K_3[Fe(C_2O_4)_3] \cdot 3H_2O$ salt (1.97 g, 4 mmol) was added in small portions to a stirred solution (50 mL) containing oxalic acid $H_2C_2O_4 \cdot 2H_2O$ (0.13 g, 1 mmol) and guanidinium carbonate $(CH_6N_3)_2CO_3$ (0.18 g, 1 mmol). After 1 h of stirring at 60 $^{\circ}C$, the mixture was cooled to room temperature and filtered. The green filtrate was allowed to evaporate in a hood at room temperature. Two weeks later, green prismatic crystals suitable for X-ray diffraction analysis were isolated. Yield: 1.58 g (82% based on $K_3[Fe(C_2O_4)_3] \cdot 3H_2O$). Anal. Calcd. for $C_{13}H_{14}Fe_2K_5N_3O_{28}$ (**1**) (%): C, 16.13; H, 1.45; N, 4.34. Found (%): C, 16.10; H, 1.47; N, 4.33. IR data (cm^{-1}): 3374 (w), 3140 (w), 2950 (s), 1712 (s), 1376 (s), 1260 (s), 887 (s), 786 (s), 503 (s), 410 (s) (Fig. S1).

2.3. Single-crystal structural determination

A suitable single crystal of the title compound was selected under an optical microscope and mounted onto a MiTeGen loop. Diffraction data were collected at 296 K on a Bruker DUO diffractometer equipped with a

Mo Incoatec Microfocus Source $I\mu S$ ($\lambda = 0.71073\ \text{\AA}$). The structure was solved with the SHELXT program [33] using intrinsic phasing solution method. The model was refined with SHELXL-2019/1 [33] using least squares minimization. All non-hydrogen atoms were refined anisotropically and all H atoms linked to N and O were either positioned geometrically or located from the electronic density difference map and refined isotropically. OLEX2 [34] and publCIF [35] programs were used for the processed crystallographic data and artwork representations. Crystal data and structure refinement details are summarized in Table 1, and selected bond lengths and angles are listed in Table 2.

2.4. Hirshfeld surface calculations

In order to gain qualitative and quantitative understandings of diverse inter-molecular interactions in salt **1**, Hirshfeld surface (HS) and their associated fingerprint plots (FP) were calculated. The Hirshfeld surface (HS) analysis is a powerful tool for visualizing interactions in molecular crystals. It's a novel means of partitioning space in molecular crystals into smooth non-overlapping molecular entities. It covers the molecule and determines the volume of space in which the electron density of the pro-molecule exceeds the density of all of the neighboring molecules [36]. HS fingerprint plots or “fingerprints” for short are convenient means of summarizing the intermolecular contacts present in crystals via a two-dimensional plot, with the decomposition of this fingerprint plot into features in order to identify particular interactions [37–39]. Based on the single-crystal structure data, the Crystal Explorer 21 program [40] was used to create the Hirshfeld surface (HS) and 2D fingerprint plots. The normalized contact distance (d_{norm}) enables the identification of the regions of particular importance to the intermolecular interactions. The 2D-fingerprint plot provides decomposition of Hirshfeld surfaces into relative contribution of different intermolecular interactions present in the crystal structures [41]. A color scale of red (shorter than Van der Waals separation)-white (equal to Van der Waals separation)-blue (longer than Van der Waals separation) is used to visualize the intermolecular contacts in the d_{norm} plot.

Table 1
Crystal data and structure refinement details for **1**.

Compound 1	
Empirical formula	$C_{13}H_{14}Fe_2K_5N_3O_{28}$
Formula weight	967.47
Temperature (K)	296
Wavelength ($\text{\AA}$)	0.71073
Crystal system	Monoclinic
Space group	$P2_1/c$
Unit cell parameters	
<i>a</i> ($\text{\AA}$)	15.6569 (7)
<i>b</i> ($\text{\AA}$)	19.9052 (9)
<i>c</i> ($\text{\AA}$)	10.5645 (5)
α , γ ($^{\circ}$)	90
β ($^{\circ}$)	109.281 (2)
<i>V</i> ($\text{\AA}^3$)	3107.8 (2)
<i>Z</i>	4
<i>D</i> _{calc} ($g\ cm^{-3}$)	2.068
μ (mm^{-1})	1.72
<i>F</i> (000)	1936.0
Crystal size (mm)	0.18 $\times$ 0.14 $\times$ 0.1
θ Range for data collection ($^{\circ}$)	1–25.7
Index ranges	–19 $< h < 19$, –24 $< k < 24$, –12 $< l < 12$
Total reflections	64,885
Unique reflections (<i>R</i> _{int})	5880 (0.070)
Refinement method	Full-matrix least squares on <i>F</i> ²
Data/restraints/parameters	5880/1/476
Goodness-of-fit (<i>G</i> _{of}) on <i>F</i> ²	1.080
<i>R</i> factor [<i>I</i> $> 2\sigma(I)$]	<i>R</i> = 0.039, <i>wR</i> = 0.103
<i>R</i> factor (all data)	<i>R</i> ₁ = 0.052, <i>wR</i> = 0.110
Max. and min. residual electron density ($e/\text{\AA}^{-3}$)	0.65/–0.74

Table 2

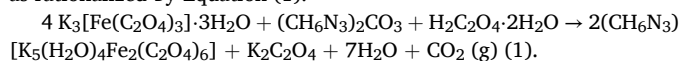
Selected bond lengths (Å) and angles (°) within the coordination spheres around the metal centers in **1**.

Fe1—O3	2.053 (2)	Fe2—O18	2.024 (2)
Fe1—O5	1.996 (2)	Fe2—O20	1.986 (2)
Fe1—O7	1.984 (2)	Fe2—O22	2.020 (2)
Fe1—O9	2.020 (2)	Fe2—O24	1.999 (2)
Fe1—O11	2.022 (2)	Fe2—O26	1.997 (2)
Fe1—O13	2.006 (2)	Fe2—O28	2.041 (2)
O5—Fe1—O3	79.09 (9)	O18—Fe2—O28	106.04 (10)
O5—Fe1—O9	90.92 (10)	O20—Fe2—O18	80.82 (9)
O5—Fe1—O11	100.02 (11)	O20—Fe2—O22	104.53 (10)
O5—Fe1—O13	89.51 (10)	O20—Fe2—O24	91.62 (10)
O9—Fe1—O3	106.54 (10)	O20—Fe2—O26	161.57 (10)
O9—Fe1—O11	161.91 (10)	O20—Fe2—O28	88.59 (10)
O13—Fe1—O3	162.77 (10)	O24—Fe2—O18	163.62 (10)

3. Results and discussion

3.1. Synthetic aspect

In order to obtain salt $(\text{CH}_6\text{N}_3)[\text{K}_5(\text{H}_2\text{O})_4\text{Fe}_2(\text{C}_2\text{O}_4)_6]$ (**1**) ($\text{CH}_6\text{N}_3^+ =$ guanidinium cation) with the expected heptanuclear $[\text{K}_5\text{Fe}_2^{\text{III}}]$ entity, an efficient method based on the ion-exchange strategy was carried out by reacting $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$ with guanidinium carbonate $(\text{CH}_6\text{N}_3)_2\text{CO}_3$ and oxalic acid $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ in a 4:1:1 M ratio in water, as rationalized by Equation (1).



It is worth noting that an aqueous solution of oxalic acid ($\text{pH} \approx 3\text{--}4$) was used in order to avoid destruction of the anionic building block $[\text{Fe}^{\text{III}}(\text{C}_2\text{O}_4)_3]^{3-}$. If a strong acid like $(\text{H}_3\text{O}^+\text{Cl}^-)$ ($\text{pH} \leq 2$) is used, the reaction is bound to fail, leading to a diversity of unexpected structures [42]. Results of elemental analysis (CHN) nicely match the chemical composition of **1**.

3.2. Structural description of **1**

X-ray diffraction analysis of $(\text{CH}_6\text{N}_3)[\text{K}_5(\text{H}_2\text{O})_4\text{Fe}_2(\text{C}_2\text{O}_4)_6]$ (**1**) shows that it crystallizes in the monoclinic space group $P2_1/c$. As depicted in Fig. 1, the asymmetric unit contains one dimetallic heptanuclear $[\text{K}_5\text{Fe}_2^{\text{III}}]$ complex anion $[\text{K}_5(\text{H}_2\text{O})_4\text{Fe}_2(\text{C}_2\text{O}_4)_6]^-$ and one guanidinium CH_6N_3^+ cation insuring the charge compensation. Relevant crystallographic parameters, selected bond lengths and angles are given in Table 2. In the anionic complex $[\text{K}_5(\text{H}_2\text{O})_4\text{Fe}_2(\text{C}_2\text{O}_4)_6]^-$, there are two crystallographically independent iron(III) ions, each with distorted

octahedral geometry comprising six O atoms from three chelating oxalato(2-) ligands. The Fe–O distances range from 1.984(2) to 2.053(2) Å and the O–Fe–O angles from 79.09(9) to 164.29(11)°. These geometric parameters are comparable with values found in the dimetallic heptanuclear $[\text{Ba}_5\text{Fe}_2^{\text{III}}]$ complex [30]. In the asymmetric unit, K1 is 9-coordinate, K2, K3 and K4 8-coordinate whereas K5 is 7-coordinate (Fig. S2). The K–O distances are in the range 2.632(3)–3.414(3) Å, matching with other reported values in M(III) oxalate based compounds [43–46]. Geometric parameters in the guanidinium CH_6N_3^+ cation show no significant difference from those of other guanidinium salts [47–51].

A crystal packing of **1** is depicted in Fig. 2. It highlights a 3-D network of $[\text{K}_5(\text{H}_2\text{O})_4\text{Fe}_2(\text{C}_2\text{O}_4)_6]^-$ complexes encapsulating guanidinium cations. Hydrogen bond lengths (Å) and bond angles (°) in **1** are listed in Table 3. The bulk structure of **1** is consolidated by a 3-D network of intermolecular $\text{O} \cdots \text{H} \cdots \text{O}$ [2.802(5) to 3.035(5) Å] and $\text{N} \cdots \text{H} \cdots \text{O}$ [2.824(5) to 3.056(4) Å] hydrogen bonds (Fig. 3) linking aqua molecules and terminal O atoms of oxalate(2-) ligands and, guanidinium cations and terminal O atoms of oxalato(2-) ligands, respectively.

Following the authoritative article on the topological classification of metal-organic frameworks (MOFs) made from multi-metal-ion units

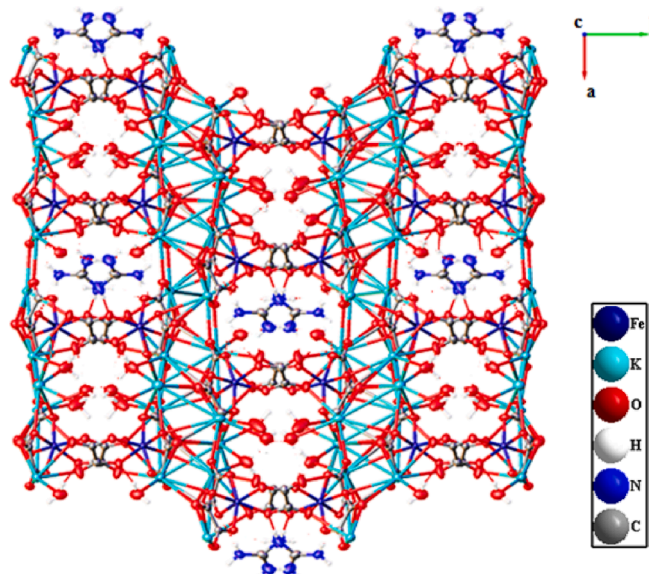


Fig. 2. Packing diagram of **1** highlighting a 3-D network of $[\text{K}_5(\text{H}_2\text{O})_4\text{Fe}_2(\text{C}_2\text{O}_4)_6]^-$ complexes encapsulating guanidinium cations.

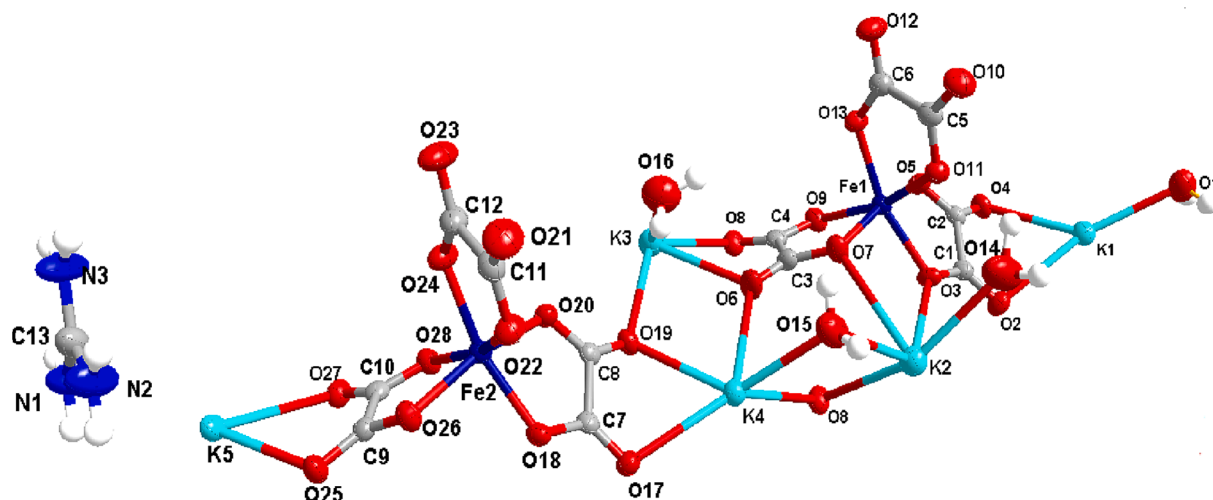


Fig. 1. Asymmetric unit of **1** with atom numbering scheme. Displacement ellipsoids are drawn at 50% probability level.

Table 3
Hydrogen bond lengths (Å) and bond angles (°) for **1**.

D—H...A	D—H	H...A	D...A	D—H...A
O1—H1A...O13 ⁱⁱⁱ	0.85	1.99	2.816	165
O14—H14A...O11	1.02	2.18	2.802	117
O14—H14B...O16 ⁱⁱⁱ	0.85	2.04	2.824	153
O16—H16A...O23 ^{iv}	0.85	2.03	2.859	164
N1—H1C...O1 ^x	0.86	1.99	2.824	162
N1—H1D...O5 ^{viii}	0.86	2.41	3.054	132
N1—H1D...O12 ^{xi}	0.86	2.41	3.023	128
N2—H2A...O18 ⁱ	0.86	2.24	3.056	159
N2—H2B...O5 ^{viii}	0.86	2.14	2.841	139
N3—H3A...O26 ^{vii}	0.86	2.09	2.919	161
N3—H3B...O22 ⁱ	0.86	1.99	2.843	172
O15—H15A...O16 ^{iv}	0.91	2.41	3.035	126
O15—H15B...O24 ⁱⁱⁱ	0.90	1.92	2.798	162

Symmetry transformations used to generate equivalent atoms (D, donor; A, acceptor):

(i) $\times, y, z+1$; (iii) $\times, y, z-1$; (iv) $\times, -y+3/2, z-1/2$; (vii) $\times, -y+3/2, z+1/2$; (viii) $\times+1, y, z+2$; (x) $\times+1, -y+3/2, z+5/2$; (xi) $\times+1, -y+3/2, z+3/2$.

[52], this polymeric structure can be considered a sheet-MOF, where metal ions and oxalates (linkers) form infinite 2D units. These MOFs have been suggested as more stable [52].

3.3. Powder XRD and thermal behavior of **1**

To check the phase purity of salt **1**, powder X-ray diffraction (PXRD) was carried out at room temperature. The PXRD pattern of the bulk sample (Fig. S3) matches well with the simulated pattern from the single-crystal data, confirming the phase purity of the target compound. The thermal stability and the decomposition process of **1** were investigated via TGA (thermogravimetric analysis) and DSC (differential scanning calorimetry) in a static air atmosphere in the temperature range 25–700 °C, as shown in Fig. 4. The first weight loss of 4.3% (calcd. 5.6%) occurring in the temperature range 130–150 °C, is an endothermic process and is due to the elimination of water molecules. The second large weight loss of 60.6% (calcd. 59.3%) occurring in the temperature range 245–625 °C, is an exothermic process and is attributed to the decomposition of the framework, the final residual product

being Fe₂O₃ [53].

3.4. Magnetic properties of **1**

The EPR spectra of powdered samples (16.3 mg, blue curve and 48.6 mg, red curve) of **1** measured at room temperature are practically similar (Fig. 5). In both cases, the anisotropy of g parameters is present along two principal axes ($g_x = 5.38$ and $g_y = 2.00$). These spectra are in line with Fe(III) complexes in an octahedral environment [30,51,54].

The temperature dependent magnetic susceptibility (χ) and inverse magnetic susceptibility (χ^{-1}) of compound **1** under zero-field-cooled (ZFC) and field-cooled (FC) conditions over 1000 Oe are depicted in Fig. 6. The match between ZFC and FC indicates that there is no magnetic anisotropy in **1**. The magnetic susceptibility χ increases gradually with decreasing temperature and increases quickly below 25 K. The temperature dependence of the inverse susceptibility χ^{-1} gives a straight line above 25 K and obeys the Curie-Weiss law $X_M^{-1} = (T-\theta)/C$ with the Curie-Weiss constants $C = 4.17 \text{ cm}^3 \text{ K mol}^{-1}$, $\theta = -9.3 \text{ K}$. The small and a negative value of θ suggests that at low temperatures, compound **1** exhibits a weak antiferromagnetic behavior [23,51,55].

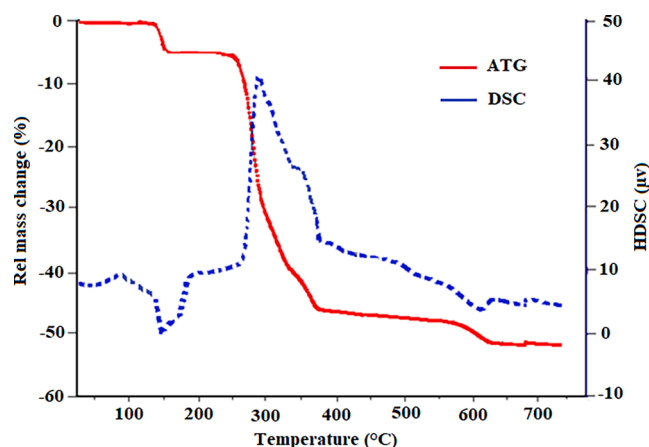


Fig. 4. TGA (red) and DSC (blue) diagrams for **1**.

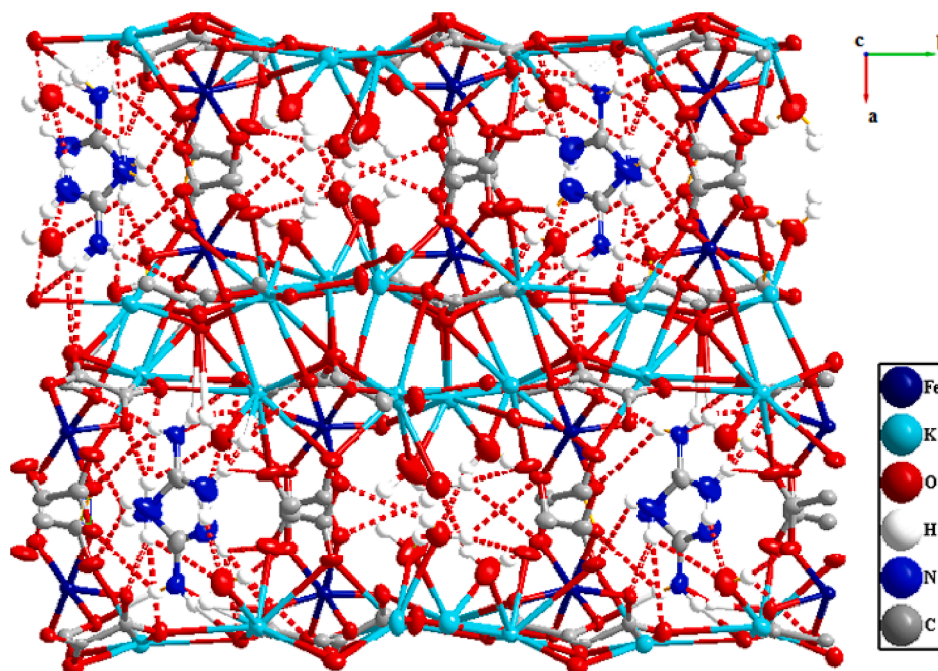


Fig. 3. Hydrogen bonding interactions (dashed lines) in **1**.

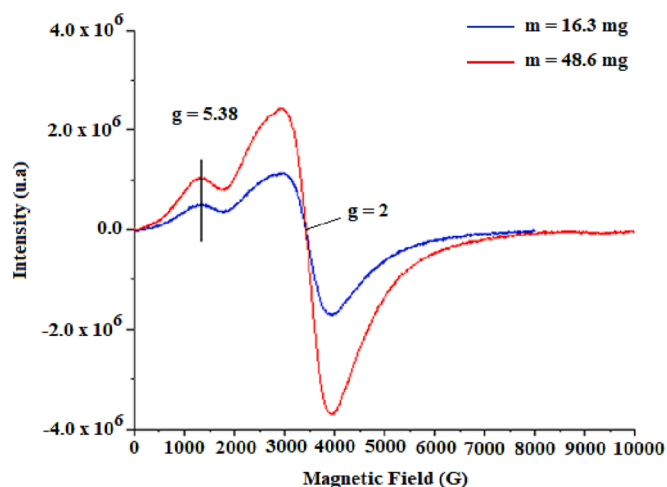
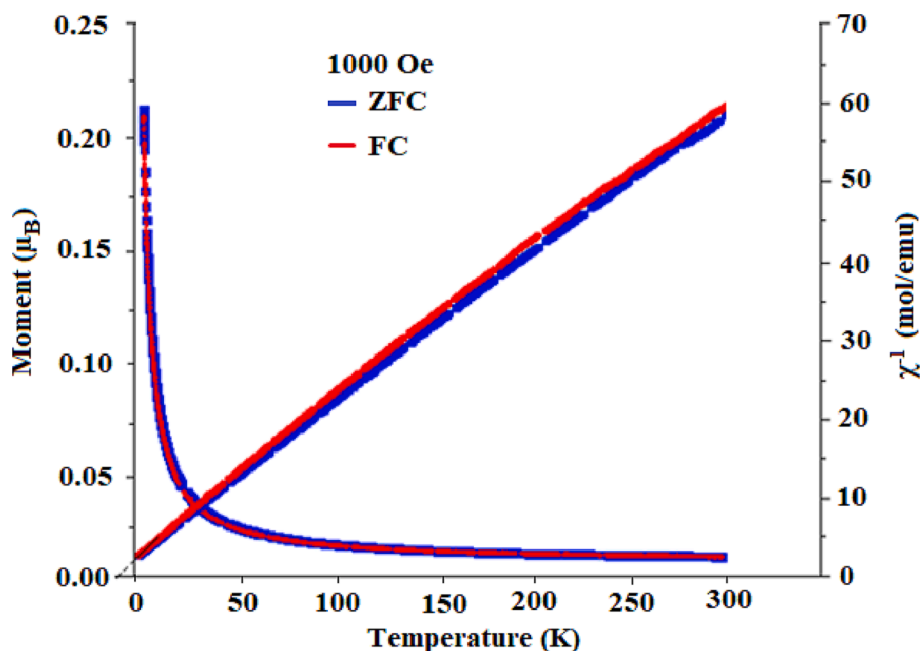
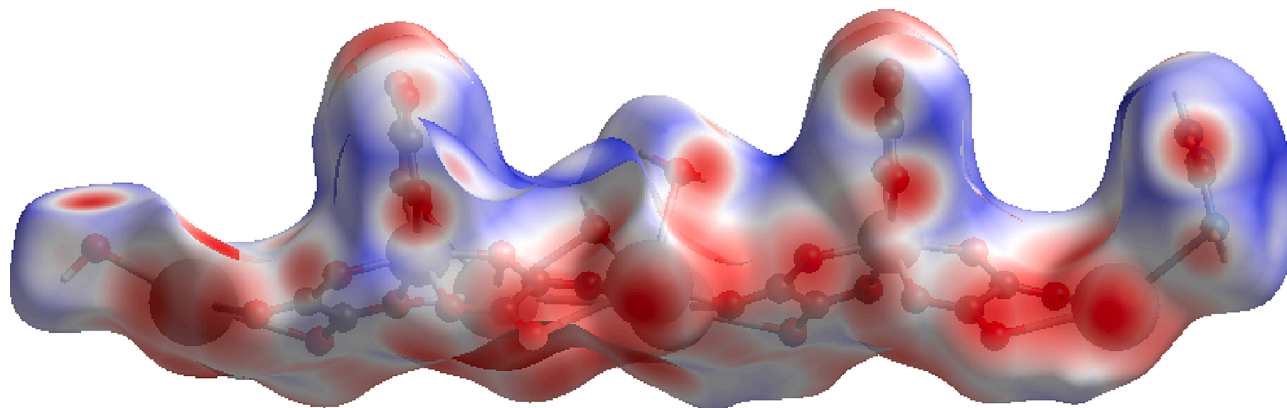


Fig. 5. EPR spectrum of compound 1.

3.5. Hirshfeld surface analysis and 2D fingerprints

A detailed analysis of the intermolecular interactions in **1** was performed by the Hirshfeld surface (HS) analysis and the equivalent two-dimensional (2D) fingerprint plots. The surface plotted over d_{norm} is shown in Fig. 7. The d_{norm} surface was mapped on a fixed color scale ranging from -0.6261 to 1.0571 . The d_{norm} values were mapped onto the Hirshfeld surface using a red-white-blue color scheme as follows: red regions represent closer contacts and a negative d_{norm} value; white areas represent contacts around the Van der Waals separation and a positive d_{norm} value. As expected, the largest red spots correspond to the presence of strong hydrogen bonds $\text{O}\cdots\text{H}\cdots\text{O}$ and $\text{N}\cdots\text{H}\cdots\text{O}$ which bring the participating atoms within a distance shorter than the sum of Van der Waals radii [56].

The 2D fingerprint plots shown in Fig. 8 depict the overlay of major different contacts over the compound's HS. The sharp spikes demonstrating $\text{O}\cdots\text{H}/\text{H}\cdots\text{O}$ appear to be the main contributor to Hirshfeld surface (36.4%). Spikes in literature are used to describe strongly hydrogen-bonded interactions [57]. The proportions of other significant intermolecular interactions for **1** are found to be $\text{All}\cdots\text{H}$ (25.7%), $\text{H}\cdots\text{H}/\text{H}\cdots\text{H}$ (7.4%) and $\text{C}\cdots\text{H}/\text{H}\cdots\text{C}$ (4.3%). These short-range interactions in

Fig. 6. Temperature dependence of the magnetic susceptibility (χ) and inverse magnetic susceptibility (χ^{-1}) of **1** under zero-field-cooled (ZFC) and field-cooled (FC).Fig. 7. Hirshfeld surface mapped over d_{norm} for **1**.

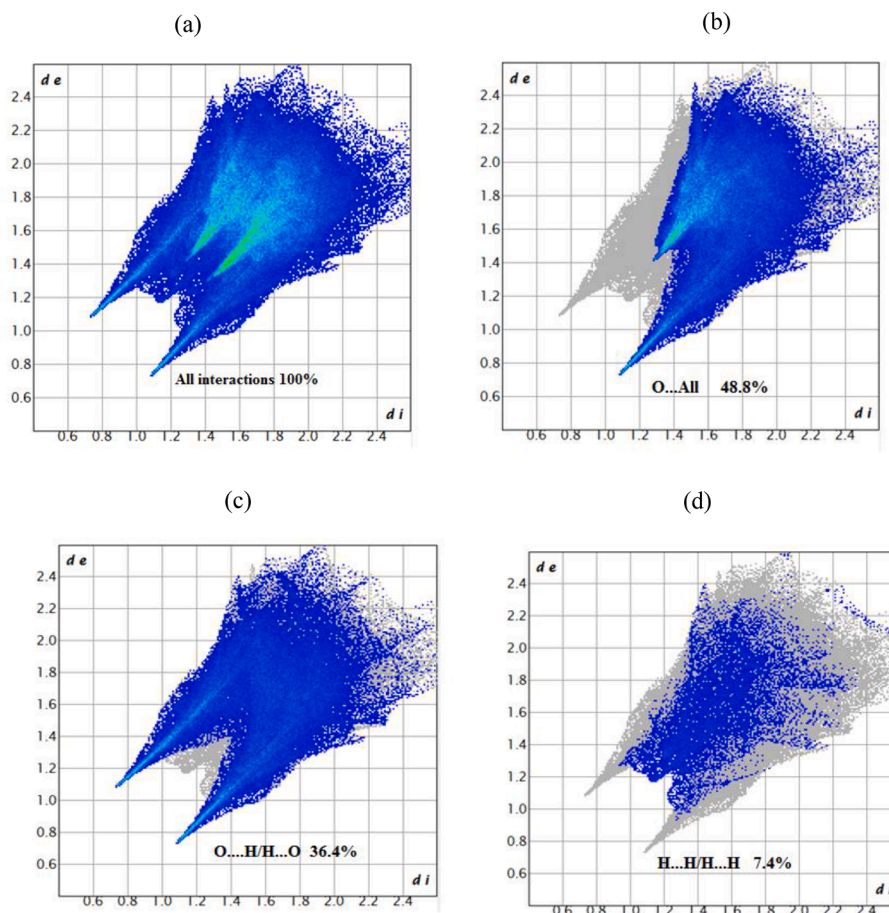


Fig. 8. Fingerprint plots and percentage contributions for major intermolecular contacts (a) All interactions, (b) O...All, (c) O...H/H...O and (d) H...H/H...H for **1**.

the structure are indispensable for the stability of crystal packing.

4. Conclusion

In summary, using an efficient method based on the ion-exchange strategy, an oxalato-bridged heterometallic heptanuclear $[\text{K}_5\text{Fe}_2^{\text{III}}]$ hybrid salt, $(\text{CH}_6\text{N}_3)[\text{K}_5(\text{H}_2\text{O})_4\text{Fe}_2(\text{C}_2\text{O}_4)_6]$ (**1**) (CH_6N_3^+ = guanidinium cation) has been synthesized and fully characterized. The structural feature of focal interest in **1** is formation of an extended anionic framework encapsulating guanidinium cations. Quantification of the various intermolecular interactions in the crystal by Hirshfeld surface analysis and the associated fingerprint plots indicated that the dominant contacts in the crystal building are $\text{H}\cdots\text{O}/\text{O}\cdots\text{H}$ contacts (36.4%). Thermal behavior investigations confirmed the anhydrous character of **1**. EPR results are in accordance with the oxidation state + 3 of the iron center in an octahedral environment. Magnetic susceptibility measurements revealed the occurrence of weak antiferromagnetic interactions between $\text{Fe}(\text{III})$ ions at low temperatures. Overall, the present finding contributes to expand this family of oxalato-bridged heterometallic polynuclear hybrid salts, and encourages us to further develop many other members for their rich structural diversity and multifunctional behavior.

CRediT authorship contribution statement

Colince Nde: Experiments, Writing original draft. **Adam Bouraima:** Investigation, Formal analysis. **Bridget N. Ndosiri:** Experiments, Writing original draft. **Idelle N. Kamga:** Experiments, Writing original draft. **Ledoux S. Pouamo:** Experiments, Investigation. **Frederic Capet:** Instrumental characterization, Software, Data curation, Draft revision

and editing. **Michel Foulon:** Instrumental characterization, Software, Data curation, Draft revision and editing. **Justin Nenwa:** Conceptualization, Methodology, Supervision, Validation, Reviewing and editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

CCDC 2234254 contains supplementary crystallographic data for **1**. These data can be obtained free of charge from the Cambridge Crystallographic Data Centre via http://www.ccdc.cam.ac.uk/data_request/cif. Supplementary data to this article can be found online at <https://doi.org/10.1016/j.poly.2023.116459>.

References

- [1] I. Matulková, J. Cihelka, M. Pojarová, K. Fejfarová, M. Dušek, P. Vaněk, J. Kroupa, R. Krupková, J. Fábry, I. Němec, *CrystEngComm* 14 (2012) 4625.
- [2] B. Zhang, Y. Zhang, Z. Wang, S. Gao, Y. Guo, F. Liu, D. Zhu, *CrystEngComm* 15 (2013) 3529.
- [3] O.M. Yaghi, M. O'Keeffe, N.W. Ockwig, H.K. Chae, M. Eddaoudi, J. Kim, *Nature* 423 (2003) 705.
- [4] A.H. Pedersen, M. Julve, J.M. Lillo, J. Cano, E.K. Brechin, *Dalton Trans.* 46 (2017) 16025.
- [5] D.B. Mitzi, S. Wang, C.A. Field, C.A. Chess, A.M. Guloy, *Science* 267 (1995) 1473.
- [6] L. Liu, Z. Bai, L. Hu, D. Wei, Z. Lin, L. Zhang, *J. Mater. Chem. C* 9 (2021) 7452.
- [7] H.Y. Shen, L. He, P.P. Shi, Q. Ye, J. Mater. Chem. C 9 (2021) 4338.
- [8] M. Clemente-Leon, E. Coronado, C. Marti-Gastald, F.M. Romero, *Chem. Soc. Rev.* 40 (2011) 473.
- [9] B. Zhang, Y. Zhang, D. Zhu, *Dalton Trans.* 41 (2012) 8509.
- [10] K. Pasinska, A. Ciupa, A. Pikul, A. Gagor, A. Pietraszko, A. Cizman, *J. Mater. Chem. C* 8 (2020) 6254.
- [11] Y.Q. Zhang, M. Li, G.C. Xu, *Appl. Organomet. Chem.* 35 (2021) e6354.
- [12] J. Laronova, B. Bombelli, J., sanchiz, O., kahn., *Inorg. Chem.* 37 (1998) 679.
- [13] C.L.F. Dazem, B.N. Ndosiri, E.N. Nfor, R. Köferstein, P. Shankhari, B.P.T. Fokwa, J. Nenwa, *J. Mol. Struct.* 1203 (2020), 127399.
- [14] L.D.D. Njiki, B.N. Ndosiri, A.N. Nana, A. Bouraima, L.S. Pouamo, F. Capet, M. Foulon, J. Nenwa, *Polyhedron* 232 (2023), 116280.
- [15] C.L.F. Dazem, F.M. Amombo Noa, J. Nenwa, L. Öhrström, *CrystEngComm* 21 (2019) 6156.
- [16] S. Decurtins, H.W. Schmalke, H.R. Oswald, A. Linden, J. Ensling, P. Gutlich, A. Hauser, *Inorg. Chem. Acta* 216 (1994) 65.
- [17] Q.H. Pan, Q. Chen, W.C. Song, T.L. Hu, X.H. Bu, *CrystEngComm* 12 (2010) 4198.
- [18] G.M. Espallargas, E. Coronado, *Chem. Soc. Rev.* 47 (2018) 533.
- [19] R. Vaidhyanathan, S.S. Iremonger, G.K.H. Shimizu, P.G. Boyd, S. Alavi, T.K. Woo, *Science* 330 (2010) 650.
- [20] C.N.R. Rao S. Natarajan R. Vaidhyanathan *Angew. Chem. Int. Ed.* 43 2004 1466.
- [21] F. Zhang, L. Hou, W. Zhang, Y. Yan, Y. Wu, R. Yang, F. Cao, Y.Y. Wang, *Dalton Trans.* 46 (2017) 15118.
- [22] N.M. ma Houga, F. Capet, J. Nenwa, G. Bebg, M. Foulon, *Acta Crystallogr. E* 71 2015 1408.
- [23] C.F.N. Nguemdzi, F. Capet, J. Ngouné, G. Bebg, M. Foulon, J. Nenwa, *J. Coord. Chem.* 71 (2018) 1484.
- [24] C.M.N. Choubau, B.N. Ndosiri, H. Vezin, C. Minaud, J.B. Orton, S.J. Coles, J. Nenwa, *J. Coord. Chem.* 74 (2021) 1209.
- [25] B. Essghaier, R. Dridi, A. Arouri, M.F. Zid, *Polyhedron* 208 (2021), 115420.
- [26] R. Dridi, B. Essghaier, S.N. Cherni, M.F. Zid, *J. Mol. Struct.* 1240 (2021), 130560.
- [27] L.S. Pouamo, C.F.N. Nguemdzi, M. Azam, D. Luong, K.A. Gibson, W. Li, E. B. Haddon, B.P.T. Fokwa, J. Nenwa, *J. Mol. Struct.* 1266 (2022), 133528.
- [28] S. Decurtins, H.W. Schmalke, R. Pellaux, P. Schneuwly, H. Hauser, *Inorg. Chem.* 35 (1996) 1451.
- [29] E. Coronado, J.R. Galan-Mascaros, C.J. Gomez-Garcia, J.M. Martinez-Agudo, *Inorg. Chem.* 40 (2001) 113.
- [30] C.M.N. Choubau, B.N. Ndosiri, H. Vezin, C. Minaud, J.B. Orton, S.J. Coles, J. Nenwa, *Polyhedron* 193 (2021), 114885.
- [31] A. Earnshaw in, *Introduction to Magnetochemistry Elsevier* 1968 1 8 10.1016/B978-1-4832-3198-3.50005-9.
- [32] Z. Liu, J. Huang, Y. Zhang, T. Liu, P. Hu, H. Liu, Q. Zheng, *Sep. Purif. Technol.* 232 (2020), 115970.
- [33] G.M. Sheldrick, *Acta Crystallogr. A* 71 (2015) 3.
- [34] O.V. Dolomanov, L.J. Bourhis, R.J. Gildea, J.A.K. Howard, H. Puschmann, *J. Appl. Crystallogr.* 42 (2009) 339.
- [35] S.P. Westrip, *J. Appl. Crystallogr.* 43 (2010) 920.
- [36] M.A. Spackman, D. Jayatilaka, *CrystEngComm* 11 (2009) 19.
- [37] M.J. Turner, J.J. McKinnon, S.K. Wolff, D.J. Grimwood, P.R. Spackman, D. Jayatilaka, M.A. Spackman, *CrystalExplorer* 17 (2017) 5.
- [38] P.H. Marek, M. Urban, I.D. Madura, *Acta Crystallogr. C* 74 (2018) 1509.
- [39] A.P. Novikov, A.A. Bezdomnikov, M.S. Grigoriev, K.E. German, *Acta Crystallogr. E* 78 (2022) 80.
- [40] P.R. Spackman, M.J. Turner, J.J. McKinnon, S.K. Wolff, D.J. Grimwood, D. Jayatilaka, M.A. Spackman, *J. Appl. Crystallogr.* 54 (2021) 1006.
- [41] M.A. Spackman, *Phys. Scr.* 87 (2013), 048103.
- [42] Y. Sun, Y. Zong, H. Ma, A. Zhang, K. Liu, D. Wang, W. Wang, L. Wang, *J. Solid State Chem.* 237 (2016) 225.
- [43] J. Nenwa, I. Kaboré, Y.A. Mbiangué, P.L. Djonwouo, P.T. Ndifon, *CSTA* 3 (2014) 67.
- [44] A.N. Nana, D. Haynes, H. Vezin, C. Minaud, P. Shankhari, B.P.T. Fokwa, J. Nenwa, *J. Mol. Struct.* 1220 (2020), 128642.
- [45] J. Nenwa, B. Gouet, P.L. Djonwouo, B.P.T. Fokwa, *Res. J. Chem. Environ.* 16 (2012) 111.
- [46] M. Bélombé, J. Nenwa, B.P.T. Fokwa, R. Dronskowski, *Acta Crystallogr. E* 62 (2006), m1400.
- [47] L. Golic, N. Bulc, *Acta Crystallogr. C* 44 (1988) 2065.
- [48] D.M. Boghaei, M.M. Najafpour, *Anal. Sci.* 23 (2007), x123.
- [49] J. Nenwa, E.D. Djomo, E.N. Nfor, P.L. Djonwouo, M. Mbarki, B.P.T. Fokwa, *Polyhedron* 99 (2015) 26.
- [50] I.N. Kamga A.N. Nana B.N. Ndosiri F. Capet M. Foulon J. Nenwa *Z. Naturforsch.* 76b (2021) 211.
- [51] C. Nde, A.N. Nana, I.N. Kamga, M. Azam, F. Capet, M. Foulon, J. Nenwa, *J. Mol. Struct.* 1263 (2022), 133145.
- [52] F.M. Amombo Noa, M. Abrahamsson, E. Ahlberg, O. Cheung, C.R. Göb, C. J. McKenzie, L. Öhrström, *Chem.* 7 (2021) 2491.
- [53] L. Wang, W. Wenqiang, D. Guo, A. Zhang, Y. Song, Y. Zhang, K. Huang, *CrystEngComm* 16 (2014) 5437.
- [54] J.T. Weisser, M.J. Nigles, M.J. Sever, J.J. Wilker, *Inorg. Chem.* 45 (2006) 7736.
- [55] M. Narsimhulu, A. Saritha, B. Raju, K.A. Hussain, *IJRSET* 4 (2015) 7548.
- [56] B.S. Yüksel, *J. Mol. Struct.* 1229 (2021), 129617.
- [57] H.A. Lamya, J. Jacques, B. Olivier, H.A. Nora, A.E. Ali, *Sci. Rep.* 9 (2019) 19745.